Belief Correction Does Not Reliably Prevent LLM Pricing Collusion

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Abstract

Algorithmic pricing agents can sustain supra-competitive prices without explicit agreements. We test whether miscalibrated beliefs about rival retaliation cause collusion, and whether exogenous belief correction reduces supra-Nash pricing. In a Calvano duopoly, an LLM focal firm reports p_retaliation each period against a scripted rival (Nash default; one-period -0.10 retaliation after a cut >0.05). A 3-arm RCT compares control (no rival-policy text), placebo (irrelevant market note), and corrected (accurate rival policy). We complete 21 episodes (7x3x120 rounds) and evaluate elevation vs Nash on rounds 71-120. Collusion is rare but real: high-price episodes in 2/7 control and 2/7 corrected runs (prices near 7 vs Nash ~4.71), and 0/7 placebo. Mean elevation: control +0.44, placebo +0.02, corrected +0.78. Belief correction does not reliably reduce supra-Nash pricing; outcomes are path-dependent.

1. Introduction

Problem. LLM pricing agents can coordinate on high prices (Fish et al., 2024), creating antitrust and AI-safety risk. Safety work needs isolatable levers.

Question (Track 1.1). Do inaccurate beliefs about rival retaliation sustain supra-Nash pricing, and does belief correction reduce elevation vs control, while a matched-length placebo does not?

Sprint contribution. Scripted retaliating rival, forecast elicitation, 3-arm RCT, calibration metrics, and a mid-scale Modal experiment (21 episodes) with plots and overall_summary.csv.

2. Related work

- Fish et al. (2024): LLM pricing collusion; prompt cues.
- Nakamura et al. (2026): private talk can introduce collusion (disabled here).
- Calvano et al. (2020): algorithmic pricing / collusion benchmarks.
- This work isolates one information lever with ground-truth rival + placebo.

3. Methodology

Environment: Calvano logit duopoly, alpha=3.2, a=2, c=1, mu=0.25. Nash ~4.713, monopoly ~6.160. Firm A = LLM (Modal support-chat); Firm B = scripted Nash-default with one-period retaliation drop 0.10 after cut >0.05.

Arms: control = none; placebo = irrelevant seasonal/shipping note; corrected = accurate rival policy.

Protocol: 120 rounds x 7 runs x 3 arms = 21 episodes. Eval window rounds 71-120. Outcomes: elevation, collusion index, Brier on cut rounds.

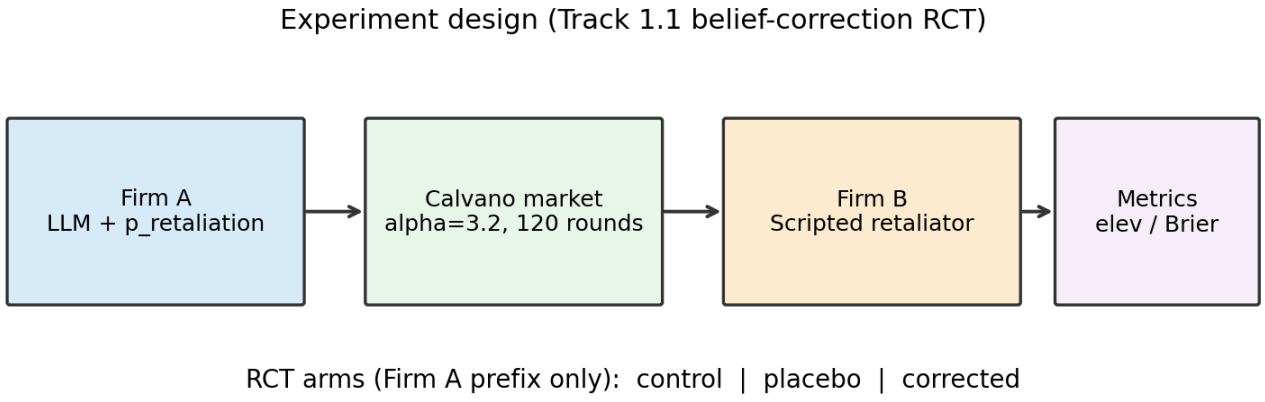


Figure 1. Experiment design.

4. Results

Table 1. Eval-window means (rounds 71-120), n=7 per arm.

Arm	Mean elev (SE)	Median elev	Mean price	High (>0.5)	Brier cuts
control	+0.440 (0.327)	+0.000	5.15	2/7	0.434
placebo	+0.024 (0.046)	-0.003	4.74	0/7	0.527
corrected	+0.784 (0.511)	-0.003	5.50	2/7	0.417

control elevations: [-0.024, 0.038, 2.287, -0.003, -0.003, 0.0, 0.787]
placebo elevations: [-0.004, -0.003, -0.103, -0.003, -0.003, 0.287, -0.001]
corrected elevations: [-0.003, -0.013, -0.003, 2.557, 2.957, -0.003, 0.0]

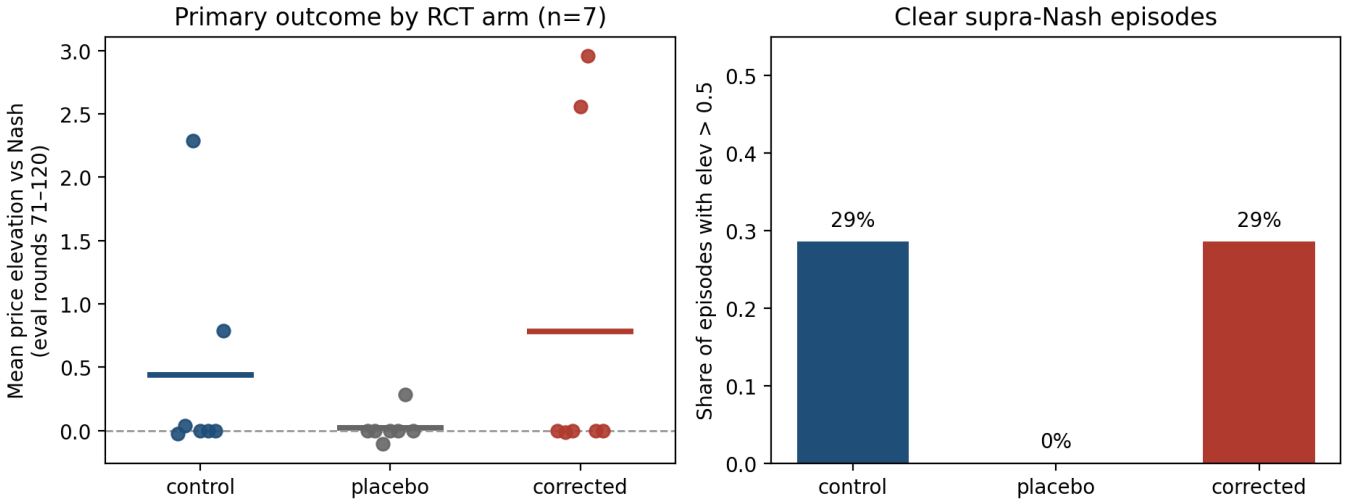


Figure 2. Elevation by arm (left) and share of clearly supra-Nash episodes (right).

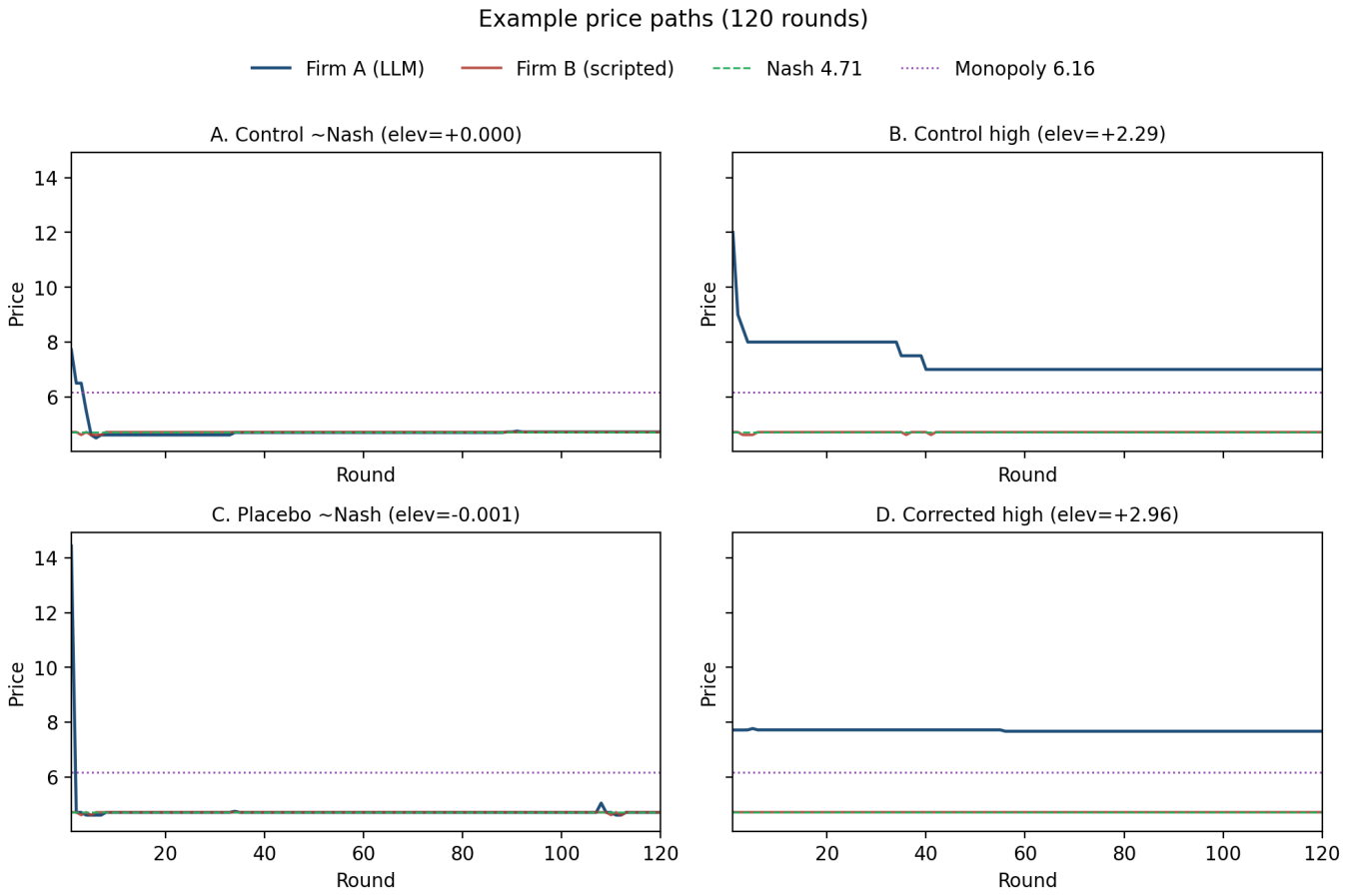


Figure 3. Example price paths: competitive convergence vs high-price lock-in.

Key findings

1. Collusion can emerge (prices ~7 > monopoly ~6.16 in some late windows).
2. Belief correction does NOT reduce mean elevation vs control (corrected mean higher due to two extremes).
3. Placebo never produced a high-price episode (0/7).
4. Path dependence dominates: medians near Nash; means outlier-driven.

5. Brier/calibration results are mixed; correction is not a consistent pricing fix.

5. Discussion

Prompted belief correction is attractive but not sufficient to prevent supra-Nash LLM pricing against a known retaliatory policy. Monitor rare high-price lock-ins, not only means or self-reported beliefs. Beliefs remain useful as measurement; mitigation needs stronger coupling to action.

Future work: longer horizons / more seeds; LLM-LLM duopoly; combine correction with communication bans or price caps.

Appendix: Limitations and Dual-Use / Ethical Considerations

Limitations: Mid-scale (120 rounds, 7 seeds); scripted rival; single model endpoint with Modal 503 cold starts; CSV n_rounds=50 is eval-window length, not episode length.

Dual-use: Detection/mitigation study for auditors/regulators. No real-market price-fixing instructions. Recommended use: stress-test LLM pricing stacks before deployment.

Preexisting vs sprint-new: Preexisting Calvano env, Experiment loop, P1 prompts, LLM agent shell. New: ScriptedRetaliatingAgent, forecast fields, RCT prefixes, belief-calibration metrics, duopoly_belief_correction.py, 21-episode run + plots.

References

1. Fish, S., et al. (2024). Algorithmic collusion with large language models.
2. Calvano, E., et al. (2020). Artificial Intelligence, Algorithmic Pricing, and Collusion. AER.
3. Nakamura, et al. (2026). Private communication and LLM collusion (sprint brief).
4. Apart Research (2026). AI Collusion Research Sprint guidelines, Track 1.1.
Data: experiments_synthetic/experiments_runs/2_agent_1/overall_summary.csv